

Bin bay screen hoarding — temporary works report

Design intent, wind check, priced build sheet and 3D model for a temporary screen hoarding round a commercial bin bay.

Prepared for	Date	Status	Reference
Darren Hopwood	4 October 2026	For review	IS-TW-BINS-001

01 What this report is — and what it is not

Read this before anything else in the pack.

This report sets out a temporary works design **intent** for a screen hoarding round the bin bay, shows the calculation every step of the way with its source, prices it, and gives a three dimensional model of it.

NOT A DESIGN. NOT FOR CONSTRUCTION. NOT A PERMIT TO LOAD.

This pack runs the calculation and shows every step. It is not the designer of record. The design and its independent check must be produced, checked and **owned** by a competent temporary works designer to BS 5975-1:2024 / BS 5975-2:2024.

Nothing is erected or loaded against this report until that designer has signed and a permit to load has been issued.

The hoarding sits inside a yard, not on the highway, so the Highways Act licensing route does not apply. Building Regulations do not govern a site hoarding either — they apply to the building. What governs is CDM 2015, the temporary works procedure, and the wind loading set out in section 03.

02 The job

What is being screened, and the arrangement proposed.

A commercial bin bay in a yard, open to the sky, with the bins standing against the building wall: three 1100 litre four wheel bins, three 240 litre bins, and a tarmac surface. Bins have to be wheeled out for collection, so the hoarding cannot simply be a closed ring.

The proposal is a **3-sided screen**: a front run across the bay with a short return at each end back to the building wall. A **1.20 m gate** in the front run keeps the bins serviceable.

ELEMENT	DESIGN INTENT
Height	2.40 m above ground
Line	6.00 m front run, two 1.20 m returns — 8.40 m of hoarding
Posts	75 × 225 mm C24, 6 no. at 2.00 m centres
Rails	100 × 75 mm C24 laid flat, three levels at 0.225 / 1.200 / 2.300 m centres

ELEMENT	DESIGN INTENT
Facing	18 mm OSB/3, on the outside face, vertical sheets
Gate	1.20 m x 2.10 m leaf, hinged and bolted
Foundations	6 no. post holes 400 mm dia x 900 mm deep, mass concrete

Why this arrangement

- **The gate matters.** A doorway does not split the hoarding into two structures (TWf2012:01 B.6), so the front run stays one continuous run of posts and rails. A closed ring would have blocked the bins.
- **Deep posts at sensible centres, not fat ones.** Stiffness goes with depth cubed. 75 × 225 C24 already carries the highest EI in the published table, so if stiffness ever bites, the answer is a deeper section, not a wider one.
- **Facing outside the posts.** The bin bay stays clear, and the boards cannot be levered off from inside.
- **Returns kept under the height.** At 1.20 m they are shorter than the 2.40 m height, so no return-corner credit is taken anywhere in the wind check. That is deliberately conservative.

03 The engineering — wind check

The hoarding is external and open to the sky, so a full wind case applies.

Method: Temporary Works Forum **TWf2012:01 (rev. August 2020)**, the simplified approach, with the wind map value taken from **BS 5975-2:2024**. The moment follows $M_o = F_w \times h/2$ — F_w is already the resultant force, so no height is applied twice.

The location is not yet confirmed, so rather than invent a wind map value the check is run across the plausible range for this kind of site and the designer locks the row once the site is named.

VB,MAP	FROM SEA	CEF	QP KN/M ²	FW KN/M	MO KNM/M	MO/POST KNM	PLANT DEPTH
22 m/s	2 km	1.174	0.362	1.305	1.75	3.50	785 mm
22 m/s	30 km	1.090	0.337	1.212	1.75	3.50	775 mm
25 m/s	2 km	1.174	0.468	1.685	2.02	4.04	835 mm
25 m/s	30 km	1.090	0.435	1.565	1.88	3.76	815 mm

Mo/post and the planting depth are at 2.00 m centres, which is the arrangement drawn. Cef from TWf Table B.1 (town, 2.5 m high); cp,net 1.50 from Table B.2 Zone B at l/h 3.5.

Adopted design intent, and how hard it is working

CHECK	RESULT
Design moment, worst row	2.02 kNm per metre run
Moment per post at 2.00 m centres	4.04 kNm
Permissible moment, 75 × 225 C24 (Table C1)	6.294 kNm
Bending utilisation	64%
Shear per post vs permissible	3.37 kN vs 17.74 kN
Post spacing the bending would allow	3.11 m — comfortably above the 2.00 m drawn
Planting depth calculated	835 mm in a 400 mm hole
Planting depth adopted	900 mm — rounded up for tolerance

The section is not what governs — robustness and stiffness are. The bending limit of 3.11 m is well beyond the 2.00 m drawn. The reason for the section is that it has to stand up to a yard full of plant and people and stay stiff while it does.

Deflection is the one check the designer must settle. Using the EI published in Table C1 it comes out at h/126 against our h/125 marker — inside it, but not by much. Using $E = 11,000 \text{ N/mm}^2$ it is h/270. Closer post centres do *not* fix it, because deflection depends on the line load only; a deeper post is the only real lever.

Every value above was cross-checked against the validated calculator that reproduces TWf2012:01 Appendix E exactly. Seven of seven values agree. The full working, step by step with its source, is in the wind check document.

04 The money

Measured quantities at building rates. Every line priced and noted.

ITEM	QUANTITY	RATE	AMOUNT
A. TIMBER			
Post 75 × 225 C24 treated, 2.40 m long	7 no	£27.50	£192.50
Rail 100 × 75 C24 treated, laid flat	27.7 m	£8.20	£227.14
Gate frame 75 × 50 C24 treated	9 m	£4.20	£37.80
B. BOARDING			
18 mm OSB/3 sheet 2440 × 1220	9 sheet	£29.50	£265.50
C. FIXINGS			
Screw 5.0 x 100, rail to post	200 no	£0.23	£46.00
Screw 4.0 x 50, board to rail and gate	100 no	£0.10	£9.50
Tee hinge, galvanised, 300 mm	1 pair	£29.00	£29.00
Sliding bolt, galvanised, 150 mm	1 no	£13.50	£13.50
D. GROUNDWORKS			
Break out and excavate 400 mm dia x 900 mm deep, through tarmac	6 hole	£48.00	£288.00
Mass concrete C20/25, placed full depth	0.71 m3	£148.00	£105.08
Tarmac reinstatement around each post	6 hole	£42.00	£252.00
E. SUNDRIES			
Sealant, washers, cutting, general consumables	1 item	£65.00	£65.00
F. LABOUR			
Carpenter, erecting	3 day	£245.00	£735.00
General labourer	1 day	£185.00	£185.00
SUMMARY		AMOUNT	
Materials, fixings and groundworks		£1,531.02	
Labour		£920.00	
Total, excluding overheads, profit and prelims		£2,451.02	
Rate per metre of hoarding (8.40 m)		£291.79	

Rates are indicative UK market rates, dated 4 October 2026, and they are there to be challenged. Drop the rate book in and the totals move. The price excludes overheads, profit and prelims; plant hire; waste removal; the temporary works design and its independent check; a CAT scan and service check; any fixing back to the building wall; and making good the wall behind the hoarding.

05 Building it

The order matters. Setting out is the whole job.

1 Set out and check the diagonals.

Front run line, posts at 2.00 m centres, returns 1.20 m back to the wall. A racked hoarding is a failed hoarding.

2 Break out and dig the holes.

6 holes, 400 mm diameter by 900 mm deep. Confirm the tarmac may be broken out and that services are clear.

3 Stand the posts, plumb and brace.

Plumb both ways, brace so they cannot move, then fill full depth with mass concrete. Do not load the posts until it has gone off.

4 Rails, then facing, then the gate.

Three rail levels on the outside face, boards vertical with joints over a rail, gate hung last on post P2 with a keep and bolt on P3.

Call off

- 75 × 225 mm C24 treated post, 2.40 m long — 7 no.
- 100 × 75 mm C24 treated rail — 27.7 m, laid flat.
- 75 × 50 C24 treated for the gate frame — 9 m.
- 18 mm OSB/3 sheets 2440 × 1220 — 9 sheets.
- Screws, one pair of tee hinges, one sliding bolt.
- Mass concrete to 6 holes — about 0.7 m³.

06 Open inputs — nothing is ordered until these close

Five things the pack cannot settle on its own.

OPEN INPUT	WHY IT MATTERS
The site	The wind map value comes from BS 5975-2 Figure 8 for the actual location. Lock the row and the designer confirms the section. The yard is assumed to be a town setting, because it sits between buildings — assumed, not proven.
The ground	Classified "average" from the tarmac surface. The geotechnical report classifies it. If it comes back poor , both the planting depth and the spacing change.
The run length	6.00 m front plus two 1.20 m returns is scaled from the photograph. Measure the bay and the drawings scale.
Breaking the tarmac	Six holes through a finished yard surface. Consent, and reinstatement, are site decisions.

Shielding

If anything that shelters the yard is claimed, it must still be standing for the whole life of the hoarding — TWf 4.2.3(c) is explicit on this.

07 The 3D model, and the documents behind this report

Scan or tap. The model works on a phone.

**3D model**

Drag to orbit, pinch to zoom. "Frame only" hides the boards so the posts and rails show.

**Full pack**

Drawings, wind check, priced build sheet and builder's briefing.

<https://darrenhopwood13-maker.github.io/tw-bin-screen-001/3d/> — the 3D model

<https://darrenhopwood13-maker.github.io/tw-bin-screen-001/> — the full pack

These links are permanent. They are hosted at the address above and will keep working, so the QR codes on this page can be sent on, printed or pinned up. Everything in this report is attached to it as well, so it stands on its own without the links.

The documents in the pack:

- **Sheet 1** — GA plan and front elevation, setting-out schedule (A3)
- **Sheet 2** — section, post foundation detail, gate leaf, general notes (A3)
- **Wind check** — every step, its value and its source, plus the cross-check (A4)
- **Priced build sheet** — PDF, Excel and CSV
- **Builder's briefing** — one page: setting out, sequence, call-off, watch-outs

The 3D model is a communication aid. It is not a structural check, not a design, and not a permit to load.

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Reference IS-TW-BINS-001, issued 4 October 2026. Prepared for Darren Hopwood. NOT A DESIGN — NOT FOR CONSTRUCTION — NOT A PERMIT TO LOAD.

Ref TW-BINS-001 Rev P01 4 October 2026 All dimensions in metres. Scale 1:50 at A3

PLAN (hoarding shown heavy; bins, wall and canopy for context only)



Set out from the left return. Total run 8.40 m, height 2.40 m.

1. A 1.20 m gate in the front run lets the bins be wheeled out on collection day. A doorway does NOT split the hoarding into two structures (TWF B.6).
2. 225 mm deep posts at 2.00 m c/c beats a fatter post stiffness goes with depth cubed.
3. Facing outside the posts keeps the bay clear and stops the boards being levered off from inside.
4. Returns are 1.20 m - less than the 2.40 m height - so no return-corner credit is taken in the wind check.

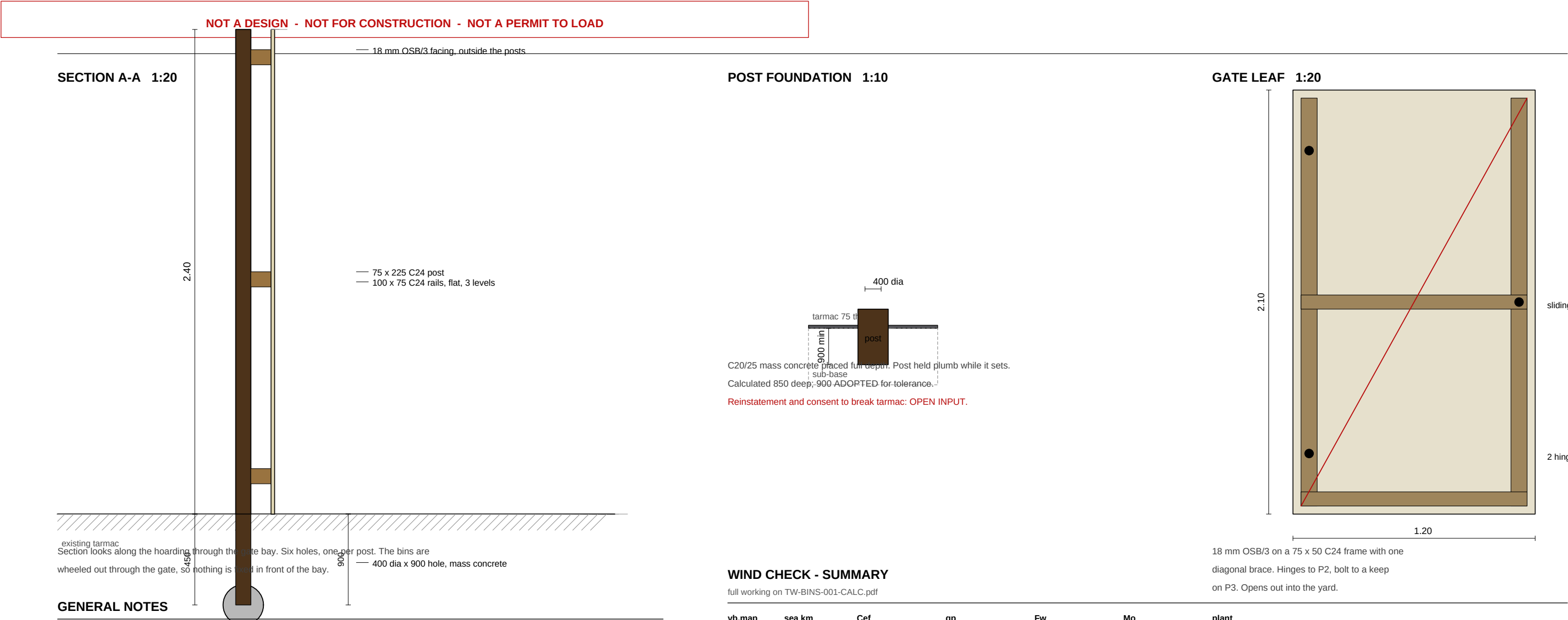
Nothing on this sheet may be built from until a competent temporary works designer has designed, checked and owned it, and a permit to load is issued.

Rev P01

NOT FOR CONSTRUCTION - NOT A DESIGN - NOT A PERMIT TO LOAD

BIN BAY SCREEN HOARDING - SECTION, FOUNDATION AND GATE DETAIL

Ref TW-BINS-001 Rev P01 4 October 2026 All dimensions in metres unless noted.



GENERAL NOTES

- NOT A DESIGN. It runs the calculation and shows every step. The design and its independent check must be produced, checked and OWNED by a competent temporary works designer to BS 5975-1:2024 / BS 5975-2:2024. Nothing is erected or loaded until a permit to load has been issued.
- Building Regulations do NOT govern a site hoarding. They apply to the building.
- EXTERNAL hoarding: the yard is open to the sky, so a full wind case applies.
- Timber C24, pressure-treated. Posts 75 x 225, rails 100 x 75 laid flat.
- Set posts first, then rails, then facing. Hang the gate last.
- No fixing into the building wall is assumed. If the returns are tied back, the fixing and its pull-out value are the designer's to specify.
- Keep the yard clear. Nothing stored against the hoarding.
- If site differs from these drawings, STOP and refer back.

WIND CHECK - SUMMARY

full working on TW-BINS-001-CALC.pdf

vb,map	sea km	Cef	qp	Fw	Mo	plant
22	2	1.174	0.362	1.305	1.75	0.785
22	30	1.090	0.337	1.212	1.75	0.775
25	2	1.174	0.468	1.685	2.02	0.835
25	30	1.090	0.435	1.565	1.88	0.815

Adopted design intent: 75 x 225 C24 at 2.00 m c/c. Mo per post 4.04 kNm against

6.294 kNm permissible - 64% used. Closer centres, not a fatter post.

The governing row is the worst of the range because THE SITE IS NOT CONFIRMED.

The designer locks one row from BS 5975-2 Figure 8 for the actual location.

SECTION + DETAILS

Rev P01

Project	Bin bay screen hoarding
Drawing	2 of 2
Reference	TW-BINS-001
Date	4 October 2026
Drawn	Banksy, for Dal Hopwood
Scale	1:20 / 1:10 at A3

NOT FOR CONSTRUCTION - NOT A DESIGN - NOT A PERMIT TO LOAD

BIN BAY SCREEN HOARDING - WIND CHECK

Ref TW-BINS-001 Rev P01 4 October 2026 | Banksy, for Dal Hopwood

NOT A DESIGN - NOT FOR CONSTRUCTION - NOT A PERMIT TO LOAD

1. WHAT THIS IS

NOT A DESIGN - NOT FOR CONSTRUCTION.

This pack runs the calculation and shows every step. It is not the designer of record.

The design AND its independent check must be produced, checked and OWNED by a competent

temporary works designer to BS 5975-1:2024 / BS 5975-2:2024.

Nothing is erected or loaded against this pack until a permit to load has been issued.

It runs the calculation and shows every step, with the source of every value. It is not competent, qualified or insured, and it cannot select the

design check category, sign a check, or issue a permit to load.

2. THE ASSET AND THE BASIS

Item	Temporary screen hoarding around a commercial bin bay (75 x 225 C24 posts, 18 mm OSB/3)
Height	2.40 m
Line	6.00 m front run, two 1.20 m returns - total 8.40 m
Post centres	2.00 m c/c, 6 posts
Exposure	EXTERNAL - the yard is open to the sky, so a full wind case applies
Site setting	town, assumed. The yard sits between buildings, so 'town' per TWf B.5
Duration	cprob = 1.0 (more than one year)
Ground	average, G = 390 kN/m2 - assumed, see open inputs

3. THE WORKING - STEP, VALUE, SOURCE

Swind = Twind x vb,map x (1 + A/1000)	25.50 m/s	BS 5975-2 simplified method; A = 20 m site altitude
cprob	1.00	TWf2012:01 B.4 - hoarding up more than a year
Cef	1.174	TWf2012:01 Table B.1 - town, 2.5 m high, 2 km from the sea
qp = 0.613 x cprob^2 x Cef x Swind^2	0.468 kN/m2	TWf2012:01 Eq B.3
l/h = 8.40 / 2.40	3.50	TOTAL_RUN 8.40 m, H 2.40 m
cp,net, Zone B, interpolated on l/h	1.50	TWf2012:01 Table B.2: 1.4 at l/h<=3, 1.8 at l/h=5
Fw = qp x Aref x cp,net x eta	1.685 kN/m run	TWf2012:01 Eq B.1, Aref = 2.40 m2/m, eta = 1.0
Mo = Fw x h/2 (max wind)	2.022 kNm/m run	TWf2012:01 s4.5 - note Fw is already the resultant, so h/2, not w h^2/2
Mo LC2 (working wind + notional 0.74 kN/m at 1.2 m)	1.752 kNm/m run	TWf2012:01 s4.2.1 and Table 2
DESIGN Mo	2.022 kNm/m run	the greater of LC1 and LC2
Mo per post at 2.00 m c/c	4.043 kNm	Mo x spacing

Permissible moment, 75 x 225 C24	6.294 kNm	TWf2012:01 Table C1
Bending utilisation	64.2 %	Mo per post / permissible
Qo per post at 2.00 m c/c	3.369 kN	Fw x spacing
Shear utilisation	19.0 %	against 17.74 kN, Table C1
Planting depth for the ground resistance	835 mm in a 400 mm hole	TWf2012:01 Eq D1/D2, ground 'average' G=390 kN/m2
Deflection, EI from Table C1 (368 kNm2)	19.0 mm = h/126	delta = w h^4 / 8EI, cantilever post
Deflection, E mean 11,000 N/mm2	8.9 mm = h/269	same formula, I = 71.2e6 mm4

Deflection is shown both ways because the two sources give different EI. Table C1's value of 368 kNm2 gives h/126 against the h/125 marker

(19.2 mm) - inside it, but not by much. With E = 11,000 N/mm2 it is h/269. The designer must decide which applies and confirm it. A deeper

post is the only fix: 75 x 225 already has the highest EI in Table C1, and closer centres do not reduce post deflection because it depends on

the line load only.

4. CROSS-CHECK AGAINST THE VALIDATED TOOL

The same inputs were run through scripts/tw_windcheck.py, the calculator that reproduces TWf2012:01 Appendix E exactly (11 of 11

published values). Agreement is what makes the figures above usable at all.

Swind m/s mine 25.500 tool 25.500 AGREES

qp kN/m2 mine 0.468 tool 0.468 AGREES

Fw kN/m mine 1.685 tool 1.685 AGREES

Mo kNm/m mine 2.022 tool 2.022 AGREES

Mo per post kNm mine 4.043 tool 4.043 AGREES

planting m mine 0.835 tool 0.835 AGREES

post spacing limit m mine 3.113 tool 3.113 AGREES

5. THE ARRANGEMENT IT SUPPORTS

- Posts 75 x 225 C24 at 2.00 m centres, 6 in total, planted 900 mm into 400 mm diameter holes in mass concrete.

- Rails 100 x 75 C24 laid flat at 0.225, 1.200, 2.300 m centres.

- Facing 18 mm OSB/3, fixed to the outside face. Gate leaf 1.20 x 2.10 m.

- Mo per post 4.04 kNm against 6.294 kNm permissible = 64% used.

6. NOT APPLICABLE, AND WHY, SO NOBODY HAS TO ASK

- Wind IS applicable. The hoarding is external and open to the sky.
- Building Regulations do NOT govern it. They apply to the building, not the hoarding.
- Highways Act 1980 s.169 does NOT apply. The hoarding is inside the yard, not on the highway.

6. OPEN INPUTS - NOTHING IS ORDERED UNTIL THESE ARE CLOSED

1. THE SITE. vb,map comes from BS 5975-2 Figure 8 for the actual location, and the table

above shows 22 and 25 m/s, coastal and inland. Confirm the site and the designer locks

the row. TOWN/CONFIRM: the yard is between buildings, so 'town' is assumed, not proven.

2. GROUND. Classification 'average' is assumed from the tarmac surface. The geotechnical

report classifies it; if it is 'poor' the planting depth and spacing both change.

3. THE RUN LENGTH. 6.00 m front + two 1.20 m returns is scaled from the photograph.

Measure the bay and the drawings scale accordingly.

4. TARMAC REINSTATEMENT and whether the surface may be broken. Six holes, 400 x 900.

5. Whether anything SHIELDS the yard that will still be there for the whole life of the

hoarding. TWf 4.2.3(c): shielding may be demolished or removed.

A wind check is not a permit to load. Nothing is erected or loaded against this until the designer signs and the permit is issued.

BIN BAY SCREEN HOARDING - PRICED BUILD SHEET

Ref TW-BINS-001 Rev P01 4 October 2026

NOT A DESIGN - NOT FOR CONSTRUCTION - NOT A PERMIT TO LOAD

Indicative UK supply rates, 4 Oct 2026, C24 treated softwood from a merchant, delivered. Excludes overheads, profit, prelims, plant hire, waste removal, design fees and the temporary works check.

Description	Qty	Unit	Rate	Amount
A. TIMBER				
Post 75 x 225 C24 treated, 2.40 m long 6 required + 1 waste. 4 to the front run, 2 at the return ends	7	no	27.50	192.50
Rail 100 x 75 C24 treated, laid flat 25.2 m measured (3 levels: 18.0 m front + 7.2 m returns) + 10% waste	27.7	m	8.20	227.14
Gate frame 75 x 50 C24 treated 2 stiles 2.10, 2 rails 1.20, 1 diagonal brace 2.40 = 9.0 m	9	m	4.20	37.80
B. BOARDING				
18 mm OSB/3 sheet 2440 x 1220 5 front run + 2 returns + 1 gate leaf + 1 waste	9	sheet	29.50	265.50
C. FIXINGS				
Screw 5.0 x 100, rail to post About 60 to the rails at 4 per connection + 112 boards to rails	200	no	0.23	46.00
Screw 4.0 x 50, board to rail and gate Boards to rails and the gate frame	100	no	0.10	9.50
Tee hinge, galvanised, 300 mm Gate leaf, hung on post P2	1	pair	29.00	29.00
Sliding bolt, galvanised, 150 mm Gate leaf, keep on post P3	1	no	13.50	13.50
D. GROUNDWORKS				
Break out and excavate 400 mm dia x 900 mm deep, through tarmac Hand dig and break out. CAT scan and service check assumed done separately	6	hole	48.00	288.00
Mass concrete C20/25, placed full depth 0.113 m3 per hole x 6 = 0.68 m3 + 5%	0.71	m3	148.00	105.08
Tarmac reinstatement around each post About 0.4 m2 per hole. CONFIRM the surface may be broken - open input	6	hole	42.00	252.00
E. SUNDRIES				
Sealant, washers, cutting, general consumables Allowance	1	item	65.00	65.00
F. LABOUR				
Carpenter, erecting 2 carpenters x 1.5 days: set out, posts, rails, facing, gate	3	day	245.00	735.00
General labourer Assisting and clearing	1	day	185.00	185.00

Materials and groundworks	1,531.02
Labour	920.00
TOTAL - excludes overheads, profit and prelims	2,451.02
Rate per metre of hoarding (8.40 m run)	291.79

WHAT THE PRICE DOES NOT INCLUDE

- Overheads, profit and prelims. Plant hire. Waste removal.
- The temporary works design and its independent check - that is the designer's fee.
- A CAT scan and service check before the holes are broken out.
- Any fixing back to the building wall, if the designer calls for one.
- Repainting or making good the wall behind the hoarding.
- Design changes if the ground comes back worse than 'average'. If it is poor, the planting depth rises and the price with it.

Rates are indicative. Replace them with the rate book before this is used commercially.

BIN BAY SCREEN HOARDING - BUILDER'S BRIEFING

Ref TW-BINS-001 Rev P01 4 October 2026 One page. Full details on the drawings.

NOT A DESIGN - NOT FOR CONSTRUCTION - NOT A PERMIT TO LOAD

NOT A DESIGN and NOT FOR CONSTRUCTION. The temporary works designer owns the design and its independent check. Do

not start until a permit to load has been issued.

WHAT YOU ARE BUILDING

A 2.40 m high screen hoarding round the bin bay: a 6.00 m front run with a 1.20 m return at each end back to the building wall. 75 x 225

C24 posts, 100 x 75 C24 rails laid flat, 18 mm OSB/3 facing on the outside, and a 1.20 x 2.10 m gate so the bins can be wheeled out.

SETTING OUT - DO THIS FIRST

- String a line for the front run. Posts at 2.00 m centres: P1 at 0.00, P2 at 2.00, P3 at 4.00, P4 at 6.00.

- Returns run 1.20 m back from each end: P5 from P1, P6 from P4. Check the diagonals before you fix anything - a racked hoarding is a

failed hoarding.

- Set out from the left return. Total run 8.40 m.

BUILD SEQUENCE

1. Break out and dig 6 holes, 400 mm diameter x 900 mm deep, one per post. Confirm the tarmac may be broken out and the services are clear.

2. Stand the posts in the holes, plumb in both directions, brace them so they cannot move, then fill with mass concrete full depth.

3. Leave the concrete to go off. Do not load the posts until it has.

4. Fix the rails to the OUTSIDE face at 0.225, 1.200, 2.300 m centres - three levels, between adjacent posts, front run and both returns.

5. Fix the 18 mm OSB/3 to the rails on the outside, vertical, joints over a rail, sheets cut to suit. Butt them tight.

6. Hang the gate last, on post P2, with a keep and sliding bolt on P3. Check it swings clear of the ground and latches.

7. Clear up. No arisings against the hoarding.

CALL OFF

- 75 x 225 C24 treated post x 2.40 m long - 7 no. (6 required)